

Sandul Henry

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EDUCATION

University of California, San Diego

B.S. in Computer Engineering – GPA: 3.9

Sept. 2023 – June 2027

San Diego, CA

- **Relevant Courses:** Linear Algebra, Analog Design, Advanced Data Structures and Object-Oriented Design, Discrete Mathematics, Calculus I/II/III, and Systems Programming

EXPERIENCE

Incoming Embedded Software Engineer

Viasat

May 2026 - Sept. 2026

San Diego, CA

Lead Embedded Software Engineer

Yonder Dynamics

Oct. 2024 – Present

San Diego, CA

- Lead undergraduate rover team by leading meetings, setting goals and standards, and recruiting and on-boarding new members. Took part in multi-disciplinary work, especially with controls and electronics.
- Wrote mission-critical code in C++ and Python to drive motor functions, arm control, and inverse kinematics.
- Utilized CAN bus between multiple high-frequency nodes, eventually adding custom CAN frames in the hardware abstraction layer, timing sending, response callback, and publishing the data.
- Integrated features with the rest of the control pipeline, including progressively adding features to the frontend (React/Redux) that integrated with platform-level systems to improve driver ease-of-use and feasibility for autonomous portions.

Research Assistant and Data Scientist

UCSD, School of Global Policy and Strategy

Dec. 2025 – Present

San Diego, CA

- Applied statistical and machine learning techniques to analyze massive datasets (500M+ records) using Python (NumPy, Pandas, PySpark)
- Performed record-matching of 7 million records in a multi-stage pipeline, going through data filtering and verification, fuzzy matching, creating and clustering sentence embeddings, and final data analysis.
- Leveraged SLURM-managed batch computing and memory-aware data pipelines to process large-scale data on remote HPC infrastructure.

PROJECTS

Full-Stack Radio Astronomy Website | hlineobs.com | *HTML/CSS/JS, Python, SQL, Linux*

Jun. 2025

- Built a remote radio astronomy system designed to observe the hydrogen spectrum; developed Python DSP pipelines to process signals and generate visualizations (e.g., PSD vs. time) onboard a hardware limited Raspberry Pi; built an automated data pipeline to email users automatically
- Designed a Flask backend with PostgreSQL to manage and store experiment requests, status updates, and act as an intermediary between the Pi and web interface; Developed backend plotting for the dynamic web interface.
- Developed an easy-to-use frontend in HTML, CSS, and JavaScript, allowing users to define and submit custom observation experiments through a web interface;

C++ Digital Signal Processing Library | [IB-DSP](#) | *C++, Signal Processing, Python, CMake*

May 2025

- Designed and implemented a C++ digital signal processing library supporting waveform generation, multiple windowing functions, and FFT-based spectral analysis.
- Optimized performance-critical DSP workloads using multithreading, improving throughput and scalability for large signal workloads.
- Built Python-based testing, validation, and visualization pipelines to verify numerical correctness and analyze behavior in both time and frequency domains, and cross-check results with established libraries.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, Javascript, HTML, CSS, SystemVerilog (HDLs), Bash Scripting, Assembly, Matlab, SQL/PostgreSQL

Frameworks: (Parallel Programming) OpenCL; (Backend) Flask, (Frontend) React

Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm

Libraries: (Python) pandas, NumPy, Matplotlib, pyLab, scikit-learn; (C++) STL, OpenCV, Qt, CMake, Make